



PRODUCT NAME

High-performance Pet Conditioner And Shampoo Formula

Description

A dual-action liquid formulation designed to cleanse and condition pet fur while maintaining a balanced pH level. This cost-effective product is specially adapted to ensure a medium viscosity for ease of application, featuring default color and fragrance to suit a variety of preferences.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	18 months
Batch Size:	10 to 100 liters
Processing Time:	3 hours
Temperature:	70°C during initial mixing, cool to 45°C
Storage Conditions:	Store at room temperature away from direct sunlight.
Equipment:	Main mixer, secondary vessel, agitators
Certification:	Not applicable

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent and base of the formulation
2	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	10.0%	Primary cleansing agent
3	Cocamidopropyl Betaine	Cocamidopropyl Betaine	8.0%	Secondary mild surfactant for cleansing and conditioning
4	Cetearyl Alcohol	Cetearyl Alcohol	7.0%	Thickening agent to adjust viscosity
5	Glycerin	Glycerin	5.0%	Humectant for moisture retention
6	Polyquaternium-7	Polyquaternium-7	4.0%	Conditioning agent to enhance fur smoothness
7	Citric Acid	Citric Acid	2.5%	pH adjuster
8	Phenoxyethanol	Phenoxyethanol	2.0%	Preservative to enhance shelf life
9	Fragrance	Parfum	1.0%	Provides a mild and pleasant scent
10	Colorant	CI 42090	0.5%	Gives a characteristic appearance
Total			100%	

Manufacturing Process

Phase 1: Phase A

- Combine Water and glycerin in the main mixer at room temperature.
- Heat to 70°C with continuous stirring.

Phase 2: Phase B

1. In a separate vessel, blend Sodium Lauryl Sulfate, Cocamidopropyl Betaine, and Cetearyl Alcohol.
2. Heat to 70°C until uniform.

Phase 3: Phase C

1. Add Phase B to Phase A slowly under constant agitation until a homogeneous mixture is achieved.
2. Cool the mixture to 45°C.

Phase 4: Phase D

1. Add Polyquaternium-7, Phenoxyethanol, Fragrance, and Colorant under gentle mixing.
2. Adjust the pH to exactly 7 using Citric Acid solution.

Required Equipment

Main mixer, secondary vessel, agitators

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store at room temperature away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Not applicable

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10 to 100 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Wet pet's fur thoroughly, apply a generous amount of product, lather well, and rinse completely. Repeat if necessary. Ensure no product enters the pet's eyes or ears.